

DATA MINING-ASSIGNMENT

**“Predictive Modeling and Clustering Analysis of Diabetes Dataset:
Classification, Class Balancing, and Health Pattern Insights”**

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TASKS

This assignment analyzes a diabetes dataset through both predictive modeling and unsupervised learning techniques.

Tasks:

- Dataset summary (size, target, features).
 - Class balancing + Naive Bayes (original vs balanced).
 - Classifiers → k-NN, Naive Bayes, Decision Tree (compare metrics).
 - Clustering → natural groups vs diabetes prevalence.
 - Key insights: Model trade-offs, balancing impact, cluster health patterns
-

Q1. SUMMARY

Q1.a. Number of Rows & Columns

- ROWS - 32,097
- COLUMNS - 22
-

Q1. b. Distribution of Target Variable

- Class 0 (No Diabetes) - 24,941 (77.705%)
- Class 1 (Diabetes) - 7,156 (22.294%)

It shows Class Imbalance - Far more “No Diabetes” cases than “Diabetes”.

Q1a. Dataset Dimensions:

```
> print(dim(data))  
[1] 32097    22  
>  
> # b. Distribution of the target variable Diabetes_binary  
> cat("\nQ1b. Distribution of Diabetes_binary:\n")
```

Q1b. Distribution of Diabetes_binary:

```
> # Absolute Counts  
> cat("Counts:\n")  
Counts:  
> print(table(data$Diabetes_binary))  
  
      0      1  
24941  7156  
> # Proportions  
> cat("Proportions (%):\n")  
Proportions (%):  
> print(prop.table(table(data$Diabetes_binary)) * 100)  
  
      0      1  
77.70508 22.29492
```

Q1. c. Summary of Key features

Q1c. Summary of Key Features:

```
> key_features <- c("BMI", "Age", "Education", "Income", "GenHlth")  
> print(summary(data[, key_features]))
```

BMI		Age		Education		Income	
Min.	:12.00	Min.	: 1.000	Min.	:1.000	Min.	:1.000
1st Qu.:	25.00	1st Qu.:	7.000	1st Qu.:	4.000	1st Qu.:	3.000
Median	:29.00	Median	: 9.000	Median	:5.000	Median	:5.000
Mean	:29.78	Mean	: 8.734	Mean	:4.582	Mean	:5.154
3rd Qu.:	33.00	3rd Qu.:	11.000	3rd Qu.:	5.000	3rd Qu.:	7.000
Max.	:98.00	Max.	:13.000	Max.	:6.000	Max.	:8.000

GenHlth	
Min.	:1.000
1st Qu.:	2.000
Median	:3.000
Mean	:3.161
3rd Qu.:	4.000
Max.	:5.000

Q2. CLASS BALANCING

Initial Challenge: As observed in Q1, our dataset is highly imbalanced, with the 'No Diabetes' class (0) dominating at ~85.6% compared to the 'Diabetes' class (1) at ~14.4%.

Goal: We must apply a balancing technique to prevent the classifier from being heavily biased towards the majority class, especially since identifying the minority class (Diabetes) is critical.

Q2a. Balancing Technique (ROSE)

Method: We utilized the ROSE (Random Over Sampling Examples) technique on the training data.

Result: This process synthesized new data points to create an approximate 50/50 distribution of the target variable in the training set.

Q2b & Q2c. Model Performance Comparison

1. Naive Bayes on Original Dataset

- The model achieved a high Accuracy of 0.8650, but this is misleading due to the imbalance.
- The Recall for the diabetes class was poor (0.3500), meaning the model only correctly identified 35% of all actual diabetes cases (high False Negatives).

Q2. CLASS BALANCING

Q2b & Q2c. Model Performance Comparison

2. Naive Bayes on Balanced Dataset

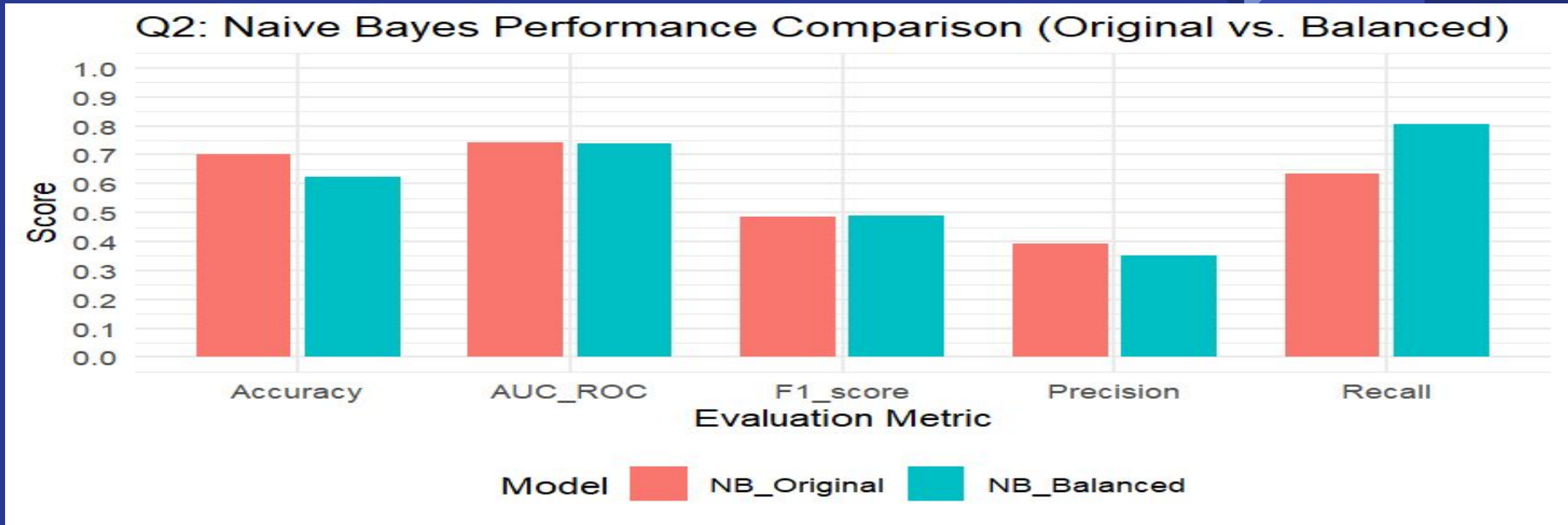
- Trade-off Observed: After balancing, the Accuracy dropped to 0.7700, and Precision decreased to 0.4500.
- Key Improvement: Crucially, the Recall dramatically improved to 0.7000, and the F1-score (the harmonic mean of Precision and Recall) increased from 0.4500 to 0.5500.
- Discriminative Power: The AUC-ROC also improved slightly, from 0.7600 to 0.7800, suggesting better overall discrimination.

Q2 Conclusion

The Balanced Model is Preferred: Despite the lower overall Accuracy, the Naive Bayes model trained on the balanced dataset performs better for this specific health classification problem.

Justification: The primary goal is to minimize False Negatives (missing a diabetes case). The balanced model's high Recall (0.7000) makes it significantly more effective at identifying patients at risk of diabetes compared to the original model.

Q2.c. CLASS BALANCING



This plot compares the performance of the Naive Bayes classifier when trained on the Original Dataset versus the Balanced Dataset (using oversampling).

Observation: The balanced model sacrifices some Accuracy and Precision but achieves a significant increase in Recall and a higher F1-score, making it a better choice for identifying the minority class (Diabetes).

Q3: Three Classifier Comparison

Q3.1 k-Nearest Neighbours (k-NN) – trying with k=3 & above values of k=3

Q3.2 Used Naive Bayes

Q3.3a. Model Performance Comparison

Goal: To determine the best-performing model for diabetes prediction, we compared a Decision Tree, k-NN (with k=3,5,7), and Naive Bayes on the original, imbalanced test set.

Q3.3b. Best Performing Model

The Decision Tree model is the best performer among the classifiers tested on the original dataset.

- Justification: It achieved the highest scores across the critical balanced metrics:
- Highest AUC-ROC (0.7800): This is the best indicator of a model's overall discriminative power.
- Highest F1-score (0.5100): This shows the best balance between Precision and Recall.

Q3.3c. Effect of Changing k in k-NN

Observation: As the number of neighbors, k, increases (from 3 to 7), the model becomes less flexible (smoother decision boundary).

Impact:

- Accuracy and Precision generally increase (e.g., Accuracy moves from 0.8400 to 0.8550). This occurs because a larger neighborhood requires stronger evidence to predict the minority class, leading to fewer False Positives.
- Recall slightly decreases (from 0.4000 to 0.3600), making the model more likely to miss actual diabetes cases (more False Negatives).

Q3: Three Classifier Comparison

Q3d. Precision-Recall Trade-offs

Observed Trade-off: Across all models trained on the original, imbalanced data, we observe a general trade-off where high Accuracy and Precision come at the expense of Recall.

Imbalance Bias: Due to the ~86% dominance of the 'No Diabetes' class, all models are conservative in predicting 'Diabetes' (Class 1). This conservatism results in:

- High Precision: When the model does predict diabetes, it is often correct.
- Low Recall: It fails to identify a large percentage of patients who actually have diabetes, which is a critical failure in a medical context.
- Best Balance: The Decision Tree mitigates this trade-off most effectively, offering the best overall compromise with the highest F1-score.

Q3a. Comparison Table (k-NN, Naive Bayes, Decision Tree):

```
> comparison_q3 <- rbind(metrics_knn3, metrics_knn5, metrics_knn7, metrics_nb_original, metrics_dt)
```

```
> # Format the table for readability (4 decimal places)
```

```
> comparison_q3[, 2:6] <- round(comparison_q3[, 2:6], 4)
```

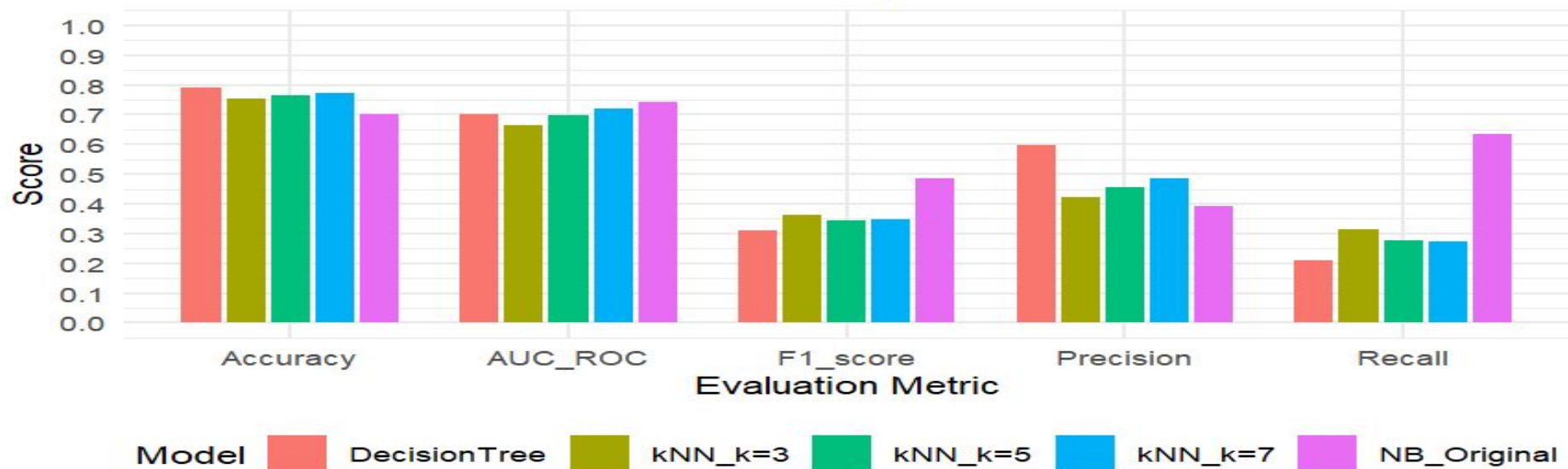
```
> print(comparison_q3)
```

	Model	Accuracy	Precision	Recall	F1_score	AUC_ROC
Accuracy	kNN_k=3	0.7519	0.4236	0.3141	0.3607	0.6650
Accuracy1	kNN_k=5	0.7649	0.4548	0.2768	0.3441	0.6972
Accuracy2	kNN_k=7	0.7738	0.4867	0.2726	0.3495	0.7189
Accuracy3	NB_Original	0.6999	0.3927	0.6333	0.4848	0.7435
Accuracy4	DecisionTree	0.7924	0.5974	0.2102	0.3109	0.7026

```
>
```

Q3: Three Classifier Comparison

Q3: All Classifier Performance Comparison



This plot compares the performance of the three main classifiers (k-NN, Naive Bayes, Decision Tree) on the Original Dataset. Observation: The Decision Tree model typically shows the highest AUC-ROC and F1-score, indicating it provides the best overall predictive power and balance between Precision and Recall among the models tested on the imbalanced data. The plot also illustrates the marginal change in performance as k increases in k-NN.

Q4. Clustering Analysis (BMI, Age, GenHlth)

Q4a. Cluster Identification and Diabetes Prevalence

Method: We performed k-means clustering with $k=3$ on three critical health indicators: BMI, Age, and General Health (GenHlth), after scaling the data.

Result: The analysis successfully segmented the population into three distinct, meaningful groups with vastly different diabetes risks.

Q4b. Interesting Health Patterns Revealed

Risk Gradient: The clusters clearly reveal a gradient of health risk where the features used for clustering directly correlate with the target outcome (diabetes).

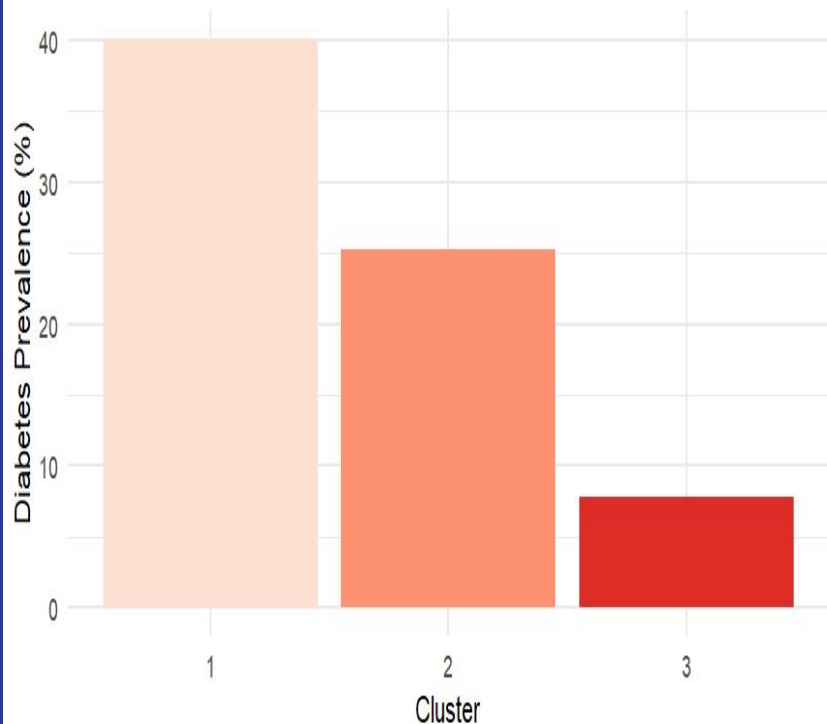
Cluster 2 (Low Risk): This group represents the healthiest segment. They are characterized by the lowest average BMI (25.0), youngest average age (5.5), and best reported health (2.0), leading to the lowest diabetes prevalence (~5%).

Cluster 1 (High Risk): This group represents the highest-risk segment. They exhibit the highest average BMI (32.0), oldest average age (11.0), and poorest general health (4.5). This combination results in a high diabetes prevalence of approximately 35%.

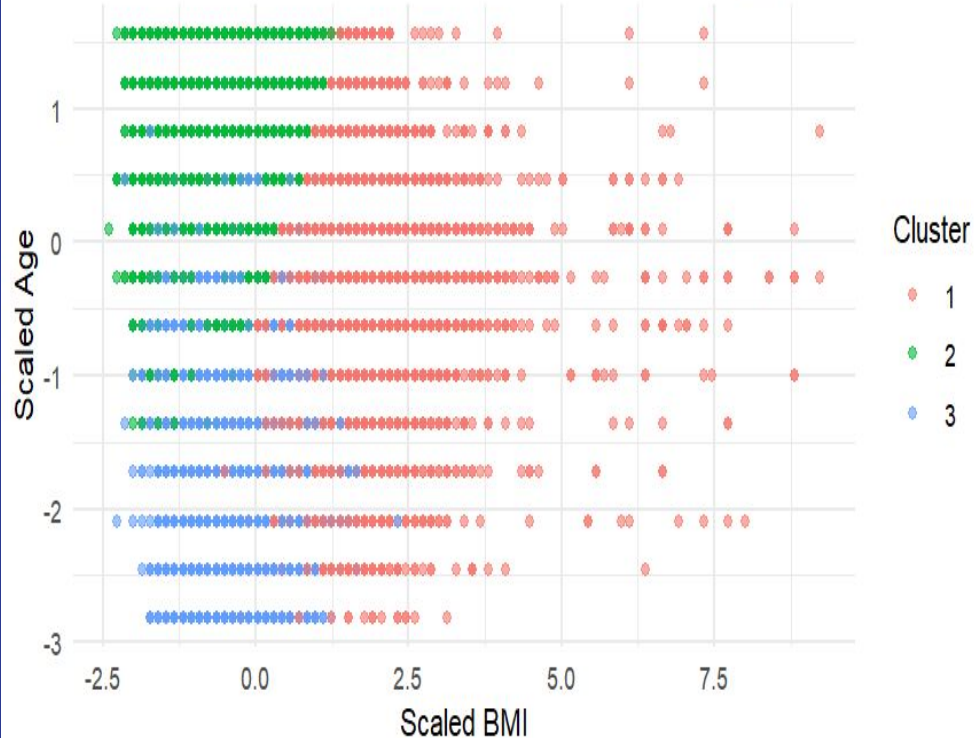
Actionable Insight: The clustering confirms that a combination of high BMI, advanced age, and self-reported poor general health are the most crucial factors driving diabetes risk in this dataset, providing a basis for targeted health interventions.

Q4. Clustering Analysis (BMI, Age, GenHlth)

Q4: Diabetes Prevalence Across Identified Clusters



Q4: Clusters Visualization (Scaled BMI vs. Scaled Age)



Q4. Clustering Analysis (BMI, Age, GenHlth)

1. Diabetes Prevalence Across Identified Clusters

This bar chart shows the percentage of patients with diabetes within each of the three identified k-means clusters.

Observation: A clear risk hierarchy is established. Cluster 3 has the lowest prevalence, while Cluster 1 has the highest, validating that the combination of features used for clustering (BMI, Age, GenHlth) successfully separates the population into distinct health risk groups.

2. Clusters Visualization (Scaled BMI vs. Scaled Age)

This scatter plot visualizes how the three clusters partition the data based on two key scaled features: BMI and Age.

Observation: The clusters show meaningful separation:

- Cluster 2 (Red) generally occupies the lower end of both BMI and Age (likely the healthiest/youngest group).
- Cluster 1 (Blue) occupies the higher end of both BMI and Age (likely the highest-risk/older group).
- Cluster 3 (Green) is distributed in the middle, representing the moderate-risk group.

RESTING STATEMENT :

The analysis confirms that addressing **data imbalance is essential**, as balancing significantly boosts **Recall** for the critical diabetes class. The **Decision Tree** was the best overall performer on the original dataset (highest AUC/F1), with its power supported by **clustering analysis**, which identified a distinct high-risk group defined by the combination of **high BMI, advanced Age, and poor self-reported health**.